

Project Synopsis

Hotel Booking Cancellation Prediction and Risk Analytics System Using Machine Learning

1. Introduction

The hotel and hospitality industry manages a large number of reservations every day. One of the major challenges faced by hotels is the cancellation of bookings, which can result in vacant rooms, inefficient inventory management, and potential revenue loss.

The proposed **Hotel Booking Cancellation Prediction and Risk Analytics System** is a Machine Learning-based project that aims to predict whether a hotel booking is likely to be cancelled. The system will analyze historical booking information such as hotel type, lead time, length of stay, number of guests, booking channel, deposit type, customer type, previous cancellations, and other relevant booking characteristics.

The project will use **Python, Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn, and Streamlit** to perform data preprocessing, exploratory data analysis, model development, evaluation, and visualization.

The project reference PDF identifies this problem as a **Binary Classification** task and proposes its application in hotels, travel agencies, online travel agencies, and other hospitality businesses.

2. Problem Statement

Hotel booking cancellations can be difficult to anticipate because they depend on several factors related to booking behavior, customer history, stay duration, booking channels, deposits, and timing.

A hotel that can identify bookings with a high probability of cancellation can take appropriate preventive measures, such as sending confirmation reminders, contacting customers, improving inventory planning, and reducing potential revenue losses.

Therefore, the problem addressed by this project is:

To develop a Machine Learning-based system that predicts whether a hotel booking is likely to be cancelled and provides a corresponding cancellation risk level based on the characteristics of the booking.

3. Aim of the Project

The primary aim of this project is to develop a **Machine Learning-based hotel booking cancellation prediction system** that can analyze historical reservation data and predict the cancellation probability of future bookings.

The system will also provide visual analytics to help understand the major factors and patterns associated with hotel booking cancellations.

4. Objectives

The main objectives of the project are:

1. To obtain and analyze a publicly available hotel booking dataset.
2. To clean and preprocess the historical booking data.
3. To perform Exploratory Data Analysis (EDA) to identify cancellation patterns.
4. To perform appropriate feature engineering on the booking data.
5. To encode categorical variables for Machine Learning.
6. To develop a **Logistic Regression** classification model as a baseline.
7. To develop a **Random Forest Classifier** for improved prediction.
8. To compare the performance of the developed models.
9. To evaluate the models using appropriate classification metrics.
10. To predict the cancellation probability of a new hotel booking.
11. To categorize bookings into Low, Medium, or High cancellation risk.
12. To develop an interactive Streamlit dashboard for prediction and analytics.

The PDF recommends data cleaning, date-based feature engineering, categorical encoding, Logistic Regression as a baseline, and tree-based models such as Random Forest as part of the Machine Learning workflow.

5. Dataset

The project will use the publicly available **Hotel Booking Demand Dataset**.

Dataset: Hotel Booking Demand

[Hotel Booking Demand Dataset — Kaggle](#)

The dataset contains historical hotel reservation records and includes information about the booking, customer, stay, and cancellation status.

Target Variable

The primary target variable is:

is_canceled

where:

0 → Booking was not cancelled

1 → Booking was cancelled

Important Features

The project will focus on relevant features such as:

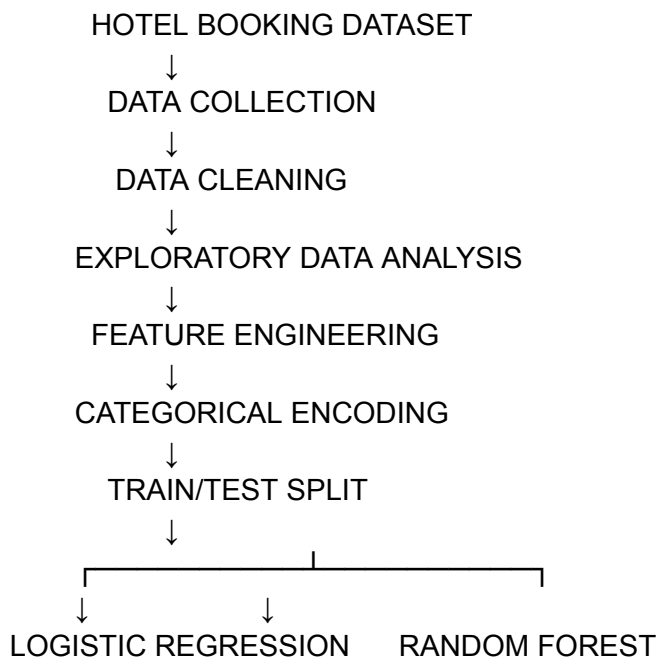
Feature	Description
hotel	Type of hotel
lead_time	Number of days between booking and arrival
arrival_date_month	Arrival month
stays_in_weekend_nights	Number of weekend nights
stays_in_week_nights	Number of weekday nights
adults	Number of adults
children	Number of children
meal	Meal plan
market_segment	Market segment
distribution_channel	Booking channel

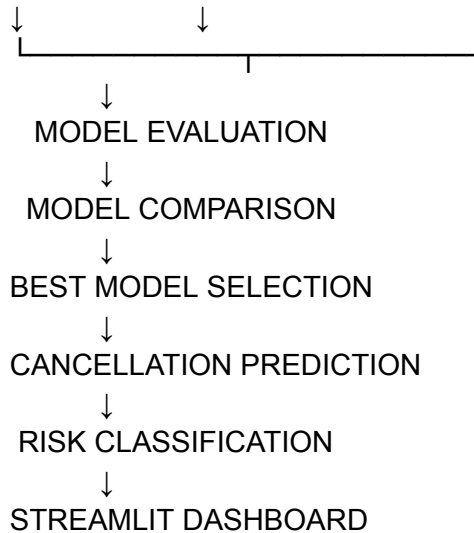
<code>previous_cancellations</code>	Previous cancelled bookings
<code>previous_bookings_not_cancelled</code>	Previous successful bookings
<code>booking_changes</code>	Number of booking modifications
<code>deposit_type</code>	Deposit type
<code>customer_type</code>	Type of customer
<code>adr</code>	Average Daily Rate
<code>total_of_special_requests</code>	Number of special requests
<code>is_cancelled</code>	Target variable

These feature categories are consistent with the project requirements in the reference PDF.

6. Proposed Methodology

The project will follow the following Machine Learning pipeline:





7. Data Preprocessing

The dataset will first be examined and cleaned before applying Machine Learning algorithms.

The preprocessing stage will include:

- Identifying missing values
- Removing duplicate records where necessary
- Handling incorrect data types
- Identifying irrelevant features
- Handling outliers where appropriate
- Encoding categorical variables
- Splitting the dataset into training and testing sets

Date-related information can also be transformed into useful features such as arrival month, arrival day, and seasonal information.

The reference PDF recommends data cleaning, date-based feature engineering, customer-level features, and categorical encoding.

8. Exploratory Data Analysis

Exploratory Data Analysis will be performed to understand the relationship between booking characteristics and cancellations.

The following visualizations will be included:

1. Cancellation Distribution

Shows the proportion of cancelled and non-cancelled bookings.

2. Cancellation by Hotel Type

Compares cancellation rates between different hotel types.

3. Cancellation by Booking Channel

Analyzes whether cancellation behavior differs between booking channels.

4. Cancellation by Deposit Type

Examines the relationship between deposit policies and cancellation behavior.

5. Cancellation by Market Segment

Identifies market segments with higher cancellation rates.

6. Cancellation by Month

Helps identify seasonal cancellation patterns.

7. Lead Time Distribution

Shows how booking lead time is related to cancellation behavior.

8. Correlation Heatmap

Helps identify relationships between numerical variables.

These visualization categories are aligned with the requirements provided in the reference PDF.

9. Machine Learning Models

Two primary classification models will be implemented.

9.1 Logistic Regression

Logistic Regression will be used as the **baseline classification model**.

Since the target variable has two outcomes:

Cancelled
Not Cancelled

Logistic Regression is suitable for establishing a simple baseline against which other models can be compared.

9.2 Random Forest Classifier

Random Forest will be used as the second classification model.

It can capture more complex relationships between different booking features and can also provide feature importance information.

The two models will be trained using the same training data and evaluated using the same evaluation metrics.

10. Model Evaluation

The performance of both models will be evaluated using:

Accuracy

Measures the percentage of total predictions that are correct.

Precision

Measures how accurately the model identifies bookings predicted to be cancelled.

Recall

Measures how many of the actual cancelled bookings are correctly identified.

F1-Score

Provides a balance between precision and recall.

ROC-AUC

Measures the model's ability to distinguish between cancelled and non-cancelled bookings.

Confusion Matrix

The confusion matrix will provide a detailed view of correct and incorrect predictions.

The PDF specifically identifies **Accuracy, Precision, Recall, F1-score, and ROC-AUC** as suitable evaluation metrics for this project.

11. Cancellation Risk Prediction

Instead of providing only a binary prediction, the system will also provide a cancellation probability and risk category.

For example:

Cancellation Probability	Risk Level
0–40%	Low
40–70%	Medium
70–100%	High

Example output:

Cancellation Probability: 78.4%

Risk Level: HIGH

Prediction: Likely to Cancel

This makes the prediction easier to understand from a business perspective.

12. Streamlit Dashboard

The final Machine Learning system will be implemented as an interactive **Streamlit web application**.

Main Dashboard

The dashboard will contain:

- Project title
- Overview of the dataset
- Key booking statistics
- Cancellation statistics
- Visualizations
- Model performance

Prediction Page

Users will be able to enter booking details such as:

Hotel Type
Lead Time
Number of Adults
Number of Children
Weekend Nights
Week Nights
Deposit Type
Market Segment
Customer Type
Previous Cancellations
Average Daily Rate
Special Requests

The system will then generate:

Cancellation Probability: 78.4%

Risk Level: HIGH

Prediction:
LIKELY TO CANCEL

13. Analytics Dashboard

The system will also provide analytical insights through graphs and statistics.

Proposed dashboard components:

Key Performance Indicators

Total Bookings

Cancellation Rate

Average Lead Time

Average Daily Rate

Visualizations

- Cancellation distribution
- Cancellation by hotel type
- Cancellation by month
- Cancellation by deposit type
- Cancellation by market segment
- Lead-time distribution
- Feature importance
- Confusion matrix
- ROC curve

The reference PDF proposes a hotel revenue dashboard that identifies risky booking segments and supports targeted confirmation and inventory/revenue planning.

14. Technology Stack

Programming Language

Python

Machine Learning

- Scikit-learn
- Logistic Regression
- Random Forest Classifier

Data Processing

- Pandas

- NumPy

Data Visualization

- Matplotlib
- Seaborn

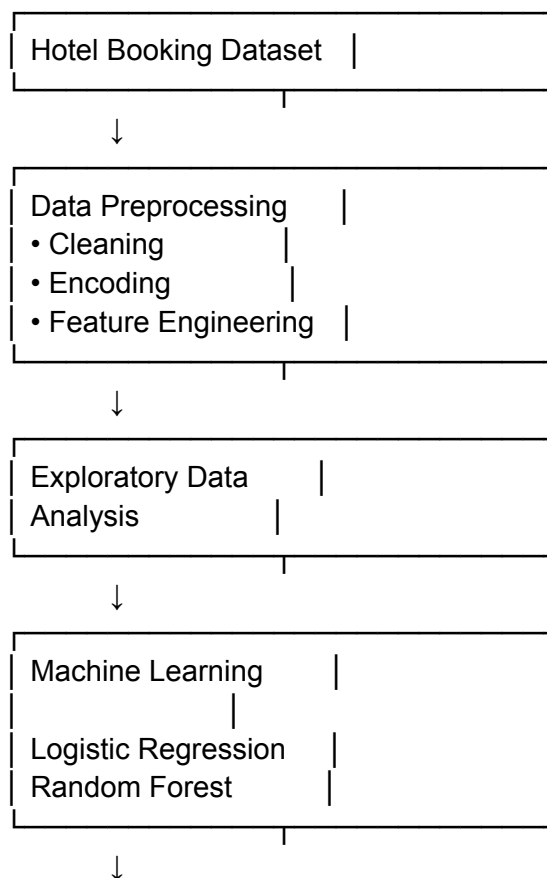
Frontend / Dashboard

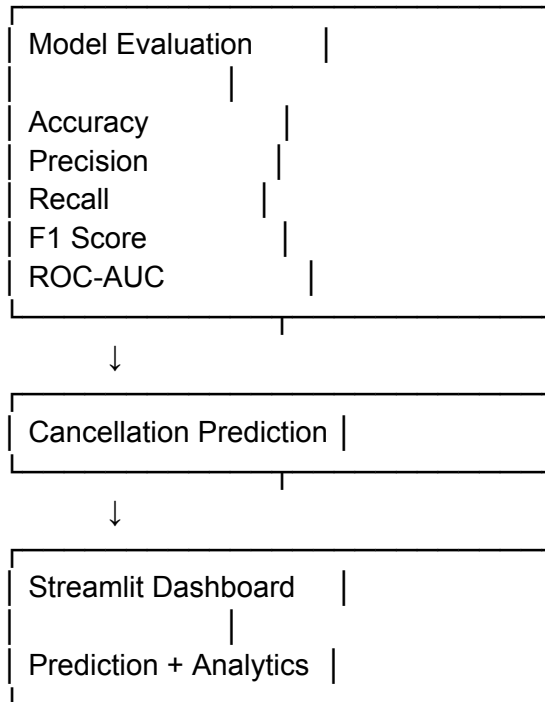
- Streamlit

Development Environment

- Jupyter Notebook / VS Code
 - Git and GitHub
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15. System Architecture





16. Expected Outcomes

The proposed system is expected to:

1. Successfully preprocess and analyze hotel booking data.
2. Identify important patterns associated with cancellations.
3. Develop classification models capable of predicting cancellations.
4. Compare Logistic Regression and Random Forest performance.
5. Identify important features influencing cancellation predictions.
6. Provide cancellation probability for individual bookings.
7. Categorize bookings into different risk levels.
8. Provide interactive visual analytics through Streamlit.
9. Help demonstrate how Machine Learning can support hotel booking and revenue-management decisions.

17. Advantages of the Proposed System

- Uses a real-world business problem.

- Uses a publicly available dataset.
 - Uses standard and understandable Machine Learning algorithms.
 - Provides both prediction and analytics.
 - Easy to visualize and demonstrate.
 - Can be deployed as an interactive Streamlit application.
 - Provides practical business insights.
 - Has scope for future expansion without making the initial project too complex.
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18. Scope of the Project

The initial version of the project will focus on **hotel booking cancellation prediction and risk analytics**.

The project will cover:

- Data preprocessing
- Exploratory Data Analysis
- Feature engineering
- Classification
- Model comparison
- Model evaluation
- Cancellation probability prediction
- Risk classification
- Interactive visualization
- Streamlit dashboard

The project will **not initially include deep learning, complex optimization, real-time hotel APIs, or advanced overbooking algorithms**. These can be considered future enhancements if required.

19. Future Enhancements

The system can be extended in the future with:

Advanced Machine Learning

- XGBoost
- LightGBM

- CatBoost

Explainable AI

SHAP or similar techniques can be added to explain why a particular booking received a high cancellation probability.

Database Integration

A database can be added to store booking records and prediction results.

Revenue Loss Prediction

The system can estimate the potential revenue loss associated with high-risk cancellations.

Real-Time Prediction API

The trained model can be exposed through an API for integration with an actual hotel management system.

Advanced Inventory Management

Cancellation predictions could be used to assist hotels in room allocation and inventory planning.

The reference PDF also identifies advanced extensions involving targeted confirmation strategies, inventory planning, revenue management, and related analytics.

20. Limitations

The system may have certain limitations:

- Prediction accuracy depends on the quality of the historical dataset.
 - Historical booking patterns may not always represent future customer behavior.
 - Unexpected events can influence cancellation behavior.
 - The initial system uses historical/static data rather than live hotel reservations.
 - The model's predictions should be considered decision-support information rather than guaranteed outcomes.
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21. Learning Outcomes

By completing this project, the developers will gain practical knowledge of:

Python

- Pandas
- NumPy
- Data preprocessing
- Data visualization
- Streamlit

Machine Learning

- Classification
- Feature engineering
- Categorical encoding
- Logistic Regression
- Random Forest
- Model evaluation
- Feature importance

Data Science

- Exploratory Data Analysis
- Data interpretation
- Visualization
- Model comparison
- Business-oriented analysis

Software Development

- Building an ML pipeline
- Developing an interactive dashboard
- Organizing a Machine Learning project
- Git/GitHub-based project management

22. Conclusion

The **Hotel Booking Cancellation Prediction and Risk Analytics System** is a Machine Learning-based solution designed to address the problem of hotel booking cancellations. The system will use historical hotel reservation data to identify patterns associated with cancellations and predict whether a new booking is likely to be cancelled.

The project will implement **Logistic Regression and Random Forest** classification models and evaluate them using standard metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC, and a Confusion Matrix. An interactive **Streamlit dashboard** will allow users to enter booking information, obtain cancellation probabilities, view risk levels, and explore hotel booking trends.

The project provides a practical application of Machine Learning to the hospitality industry while maintaining a **manageable technical scope** suitable for an academic project. It can also be extended in the future with advanced models, explainable AI, revenue-loss estimation, databases, and real-time prediction capabilities.